

# Michael Chen

+1 613-666-8666 | mchen522@uwo.ca | LinkedIn | Web | GitHub

## Education

**Master of Data Analytics(Specializing in AI)** *Western University, London, Canada* Expected 08/2026

- Relevant Coursework: Artificial Intelligence, Reinforcement Learning, Structured/Unstructured Databases

**HBSc in Mathematics & Computer Science** *University of Toronto SG, Toronto, Canada* 09/2020 – 09/2024

- Relevant Coursework: Neural Networks, Image Understanding, Data Analysis, Mathematical Modeling

## Skills

- **Programming Languages:** Python, Java, SQL, JavaScript/TypeScript, R
- **AI/ML:** PyTorch, Transformers, LLMs (GPT-4, Claude, Llama), RAG, Agent, RAGAS
- **Data Engineering:** PySpark, Docker, Azure, MongoDB, ETL/ELT Pipelines
- **Analytics:** Statistical Modeling, Computer Vision (OpenCV), Power BI

## Experience

**Data Analyst & Developer** | *Trading Card Business, Toronto, Canada* 01/2022 – Present

- Architected pcomputer vision grading system using Python and OpenCV, achieving 90% accuracy and reducing card evaluation time by 98% (5 minutes to 10 seconds per card)
- Saved 15 hours monthly by building automated data pipeline integrating web scraping and real-time alerting system that identified profitable arbitrage opportunities
- Increased ROI by 200% by analyzing and comparing historical transactions to optimize buying decisions and minimize low-grade inventory risk

**AI Solutions Developer** | *Golf Town, Burnaby, Canada* 02/2025 – 07/2025

- Developing enterprise RAG chatbot using LangChain and Llama 3.1 to handle 10+ daily customer inquiries, reducing knowledge retrieval time by 90% (5 minutes to 30 seconds)
- Analyzing ball-flight and customer fitting data using statistical methods, integrating insights into RAG pipeline to enhance recommendation accuracy and customer satisfaction
- Improving recommendation quality by comparing processed customer fitting data to 500+ equipment specifications via RAG system, reducing advisor bias and increasing compatibility

## Projects

**CRM Integration Platform** | *Vessl Prosthetics Inc Industrial Project, Python* 11/2025

- Built production-grade REST API middleware synchronizing 10K+ contacts between Fibery and HubSpot platforms, eliminating 25+ hours of manual data entry per week across business (CRM) systems
- Engineered intelligent data pipeline with Smart Merge algorithm performing field-level comparison and selective updates. Automated ETL workflows with custom data transformation logic (name parsing, field mapping, validation) reducing manual intervention by 100% and preventing data loss in multi-team collaborative environments

**Hybrid Multi-Agent RAG Framework** | *LangGraph + LlamaIndex* 09/2025

- Achieved enterprise-grade RAGAS validation scores (faithfulness 0.95, answer relevancy 0.92, context precision 0.89) by developing production-ready multi-agent system with LangGraph orchestration and LlamaIndex RAG, improving answer correctness from 0.60 to 0.95
- Optimized RAG pipeline through dimensional embedding tuning, hybrid BM25+dense retrieval, and advanced prompt engineering (role-based prompts, Recursive prompting, structured outputs), achieving 35% performance improvement
- Engineered stateful conversation system with contextual memory, conversation history tracking, and hierarchical knowledge sharing enabling agents to maintain context across multi-turn interaction

**Piano Melody Generator** | *Project Lead, PyTorch* 12/2023

- Led and mentored 4-person development team, conducting weekly strategy sessions and implementing collaborative workflows that improved team productivity by 30% and ensured timely project completion
- Drove consensus on technical decisions by facilitating data-driven discussions around model performance metrics, leading team to select optimal 2-layer LSTM architecture based on comprehensive user feedback (6.78/10 rating from 50+ evaluators)

## CERTIFICATIONS

**Microsoft Certified: Azure Fundamentals (AZ-900)** 09/2024

- Certification ID: 1100115363